

=== GNB (1780332485) ===

```
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#

# This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
# This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth
# To run the srsRAN Project gNB with this config, use the following command:
#   sudo ./gnb -c gnb_zmq.yaml

cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1      # SST 1 es comúnmente usado para eMBB (Enhanced Mobile Broadband)
    inactivity_timer: 60    # Sets the UE/PDU Session/DRB inactivity timer to 60 seconds (1 minute).

ru_sdr:
  device_driver: zmq        # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
  srate: 11.52             # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75              # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75              # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500         # ARFCN of the downlink carrier (center frequency).
  band: 3                  # The NR band.
  channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 15           # Subcarrier spacing in kHz used for data.
  plmn: "00101"           # PLMN broadcasted by the gNB.
  tac: 7                   # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0         # Set search space zero index to match srsUE capabilities
      coresets0_index: 6   # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common     # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1  # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64       # Sets PDSCH MCS to 64 QAM
  pusch:
```

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mcs_table: qam64                                # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log                        # Path of the log file.
  all_level: info                              # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false                            # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap              # Path where the MAC PCAP is stored.
  ngap_enable: true                            # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap            # Path where the NGAP PCAP is stored.
  e2ap_enable: true                            # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap      # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:
#  enable_du_e2: true                          # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true                      # Enable KPM service module
#  e2sm_rc_enabled: true                      # Enable RC service module
#  addr: 127.0.0.1                            # RIC IP address
#  bind_addr: 127.0.0.100                    # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required if
#  port: 36421                                # RIC port

metrics:
  layers:
    enable_rlc: true                          # Enable RLC metrics
    enable_sched: true                        # Enable DU scheduler metrics
  periodicity:
    du_report_period: 400                    # Set DU statistics report period to 400ms

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=== UE (1780332485) ===

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#

[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 52
nof_prb = 52

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 001010000000001
imei = 353490069873319

[rrc]
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```
release = 15  
ue_category = 4
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```
[nas]  
apn = srsapn  
apn_protocol = ipv4
```

```
[gw]  
netns = ue1  
ip_devname = tun_srsue  
ip_netmask = 255.255.255.0
```

```
[gui]  
enable = false
```

=== DOCKER (1780332485) ===

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#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  env_file:
    - ${OPEN5GS_ENV_FILE:-open5gs/open5gs.env}
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
  # - "38412:38412/sctp"
  # - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
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```
gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  # privileged mode is required only for accessing usb devices
  privileged: true
  # Extra capabilities always required
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    # Access USB to use some SDRs
```

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- /dev/bus/usb/:/dev/bus/usb/
# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.1}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ${GNB_CONFIG_PATH:-./gnb.yaml} # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0
        metrics:
          autostart_stdout_metrics: true
          enable_json: true
        remote_control:
          bind_addr: 0.0.0.0
          enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780332485) ===

1. PLMN (MCC 001, MNC 01, TAC 7) y IMSI (001010000000001) son coherentes (Regla 1).
2. dl_arfcn 368500 para Banda 3 es válido (Regla 2).
3. common_scs 15 kHz es coherente (Regla 3).
4. Puertos ZMQ 2000/2001 cruzados entre gNB y UE (Regla 4).
5. bind_addr configurado según la Regla 5 para gNB standalone y docker-compose.
6. Bloque ru_sdr copiado exactamente del CAG (Regla 6).
7. SST 1 para eMBB añadido al gNB, valor común para este tipo de servicio.
8. inactivity_timer del gNB configurado a 60 segundos.
9. Métricas RLC y Scheduler DU activadas con periodo de 400ms.
10. PCAP NGAP habilitado en gNB.